

Highlights

Continuous Trajectory Tracking for Non-Spherical Wrist Manipulator via Coupled-Reward-Driven Differential Inverse Kinematics

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- [Coupled](#) min-operator, progress, and [joint velocity penalty](#) reward enables stable IK.
- Min-operator coupling prevents reward exploitation across pose objectives.
- Policy trained only on static poses generalizes zero-shot to four trajectories.
- [Lowest mean smoothness index of all methods on circle, lemniscate, and Lissajous.](#)

Continuous Trajectory Tracking for Non-Spherical Wrist Manipulator via Coupled-Reward-Driven Differential Inverse Kinematics

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Abstract

Continuous trajectory tracking requires differential inverse kinematics (IK) at every control timestep; for non-spherical wrist manipulators, kinematic coupling renders this a nonlinear problem. Numerical solvers that resolve way-points independently [can produce joint-space discontinuities; seeding from the previous solution, joint velocity limiting, and trajectory smoothing reduce but do not remove this behavior](#). We address this through a coupled reward function, combining a min-operator inverse-quadratic pose reward, asymmetric progress shaping, and an exponential [joint velocity penalty](#), that enforces simultaneous position-orientation convergence while suppressing velocity discontinuities, applied here to a Proximal Policy Optimization (PPO) controller mapping a 19-dimensional observation directly to joint velocity commands. The policy is trained in Isaac Lab on 1.5 M joint-limit-aware reachable workspace poses, with no trajectory supervision. The proposed method achieves competitive position and orientation accuracy in simulation against baselines spanning a numerical IK solver, classical velocity-level controllers, and contemporary RL methods, and outperforms the alternative RL baselines [\(TRPO-D, trained on the identical reward, and SAC-S\)](#). Zero-shot sim-to-real transfer to the Kinova Gen3 Lite at 20 Hz closed-loop frequency achieves a position RMSE of 6.3 mm and orientation RMSE of 0.032 rad on circular trajectory tracking. The policy generalizes without retraining to square, lemniscate, and 3-D Lissajous trajectories, demonstrating that a velocity-level differential IK mapping learned purely on static pose reaching transfers directly to continuous trajectory tracking at deployment, [demon-](#)

strated on the four trajectory geometries evaluated here.

Keywords: Differential Inverse Kinematics, Trajectory Tracking, Non-Spherical Wrist Manipulator, Reward Shaping, Reinforcement Learning, Sim-to-Real Transfer.

1. Introduction

Robotic manipulators are indispensable to modern industrial automation, serving as the backbone for tasks ranging from precision assembly to complex material handling. To meet the demands of rapid advancements in the field, contemporary systems must provide not only mechanical precision but also intelligent, computationally efficient manipulation in real-time [1]. Traditionally, spherical wrist designs have dominated the industry due to their kinematic decoupling, which facilitates independent, closed-form analytical solutions for position and orientation [2]. While these designs have long been the industrial standard, they impose significant structural limitations on modern applications [3]. The geometric constraint requiring the final three axes to intersect at a common point, often compromises mechanical stiffness and restricts the integration of high-torque motors within compact wrist assemblies [4]. Consequently, non-spherical wrist manipulators have gained prominence. By introducing offsets between the last three axes as shown in Fig. 1, these configurations achieve superior structural integrity and higher payload-to-weight ratios [5]. However, this violation of kinematic decoupling renders analytical inverse kinematics (IK) intractable, necessitating a reliance on complex numerical or data-driven solvers.

Modern industrial practices require far beyond standard pick and place operations [6]. Tasks such as continuous inspection and dynamic human-robot collaborative handoff increasingly require manipulators to track end-effector trajectories whose target poses are determined online and cannot be fully enumerated or precomputed in advance, in contrast to fixed, repeated motion profiles for which offline trajectory planning remains both sufficient and preferable [7]. These operations require the manipulator to maintain a precise end-effector pose along a continuous Cartesian path, requiring joint velocity resolution at the hardware’s closed-loop control frequency [8, 9]. However, for non-spherical wrist manipulators, this requirement introduces a fundamental kinematic bottleneck. The offset joint structure creates a unified, non-linear system where position and orientation are deeply coupled,

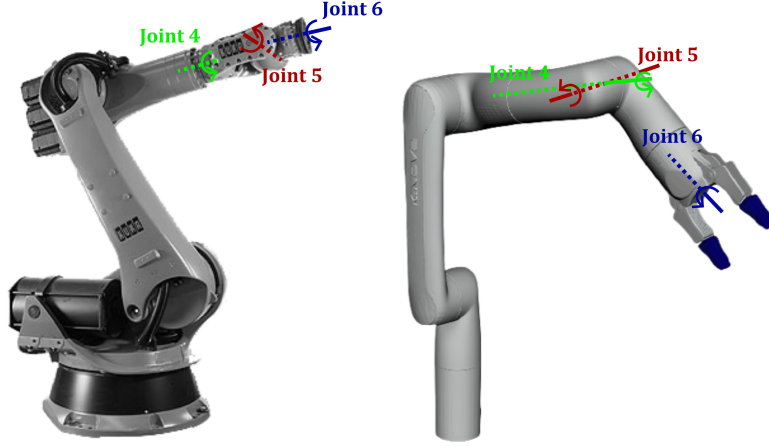


Figure 1: (a) KUKA KR-240 Manipulator with Spherical Wrist (b) Kinova Gen3 Lite Manipulator with Non-spherical Wrist

requiring the inverse kinematics to be solved as a single optimization at every control timestep.

In this work, we present an end-to-end PPO-based differential inverse kinematics controller that directly maps instantaneous Cartesian pose errors to joint velocity commands, bypassing any dependence on numerical solvers or explicit Jacobian computation. A novel coupled reward function combining a min-operator pose reward, asymmetric progress shaping, and exponential [joint velocity penalty](#) enforces simultaneous position-orientation convergence while producing smooth, hardware-safe velocity commands. Sim-to-real transfer is validated on physical Kinova Gen3 Lite hardware at 20 Hz.

The remainder of this paper is organized as follows: Section 2 surveys existing works on numerical IK solvers, velocity-level controllers, and learning-based approaches. Section 3 details the proposed methodology, including the simulation environment, reward function design, and ablation studies validating each reward component. Section 4 presents quantitative and qualitative comparisons of the proposed method against classical and learning-based baselines on simulation and our method on real hardware. Finally, Section 5 summarizes the conclusions drawn from this work and outlines directions for future research. Our contributions are summarized as follows:

- We design a coupled reward architecture, combining min-operator pose

coupling, asymmetric progress shaping, and an exponential [joint velocity](#) penalty that resolves simultaneous position-orientation convergence for the kinematic coupling inherent to non-spherical wrist manipulators, without manual inter-objective weighting or staged training curricula. This reward formulation, rather than the choice of underlying optimizer, constitutes the core engineering contribution of this work. [The architecture is geometry-agnostic and broadly applicable. We focus on non-spherical wrists because their inherent kinematic coupling prevents analytical decoupling. Unlike spherical wrists, which admit closed-form solutions, non-spherical configurations require the reward formulation to explicitly enforce simultaneous convergence.](#)

- [We show, through comparisons against alternative on-policy and off-policy algorithms \(TRPO, SAC\) trained on the identical dense reward, and against PPO trained on a sparse reward, that the proposed reward architecture is necessary for the controller’s accuracy-smoothness trade-off, and that, among the optimizers tested, Proximal Policy Optimization \(PPO\) is the most robust to the conflicting gradients it introduces, making it an empirically justified choice of optimizer.](#)
- We demonstrate that a controller trained purely on static, joint-limit-aware reachable-pose samples with no trajectory supervision, dynamic model, or task-specific retraining generalizes zero-shot to four continuous trajectory geometries (circle, square, lemniscate, 3-D Lissajous) in both simulation and physical hardware, offering a reactive, online alternative to precomputed offline IK pipelines for applications where target poses cannot be fully enumerated in advance.

2. Related Work

To address the general inverse kinematics problem, several classical methods like numerical, analytical, combination of both and heuristic frameworks have been used in literature [10]. Iterative solvers like TRAC-IK [11] are widely used for their system-agnostic ability to handle joint limits and multiple constraints. However, even with a previous joint-state seed and the Distance parameter for solution selection, TRAC-IK remains fundamentally trajectory-unaware, each waypoint is resolved as an independent optimization with no inter-waypoint coupling, leading to configuration-branch switching and joint-space discontinuities during continuous trajectory tracking.

88 Practical implementations mitigate this behavior in three ways, each of which
 89 carries a trade-off rather than removing the underlying problem. First, seed-
 90 ing every solve with the previous solution keeps consecutive solutions on the
 91 same branch while the competing branches remain well separated; on a fully
 92 coupled 6-DOF problem this cannot be guaranteed along an arbitrary path,
 93 since branches approach one another near singular configurations and a lo-
 94 cal solver may converge to a different branch, or fail to converge, from its
 95 seed [11, 12]. Consistent with this, the seeded TRAC-IK baseline evaluated
 96 in Section 4.1 attains the lowest position error of any method on the planar
 97 curves, yet has the largest smoothness index of any method and degrades
 98 markedly on the 3-D Lissajous trajectory. Second, limiting the commanded
 99 joint velocity bounds the magnitude of any single discontinuity but does
 100 not prevent the branch change itself: clipping joint velocities individually
 101 distorts the path, whereas uniformly time-scaling the motion preserves the
 102 geometric path at the cost of a tracking lag that grows with the reference
 103 speed [13]. Third, trajectory smoothing and lookahead filtering are effec-
 104 tive for pre-planned motion profiles whose full path is known in advance; for
 105 targets determined online, only causal smoothing or online trajectory gen-
 106 eration is available [14, 15], which bounds velocity, acceleration, and jerk
 107 but likewise smooths a branch change rather than preventing it, and adds a
 108 delay relative to the reference. The statements made in this work therefore
 109 concern the behavior that remains after these mitigations are applied, rather
 110 than numerical IK in an unmitigated form.

111 Similarly, heuristic methods like Forward And Backward Reaching IK
 112 (FABRIK) [16] are computationally efficient but limited to position-reaching
 113 tasks and lack temporal awareness. Additionally, the analytical closed-form
 114 solution in [17] reduces the robot’s complex kinematics to a single 16th-
 115 degree polynomial to compute all viable joint configurations for a given pose.
 116 Without strict nearest-joint constraints, continuous sub-millisecond tracking
 117 causes the solver to jump erratically between distinct configurations. This
 118 branch-jumping behavior is more in non-spherical wrists. While spherical
 119 wrists decouple IK into independent position and orientation subproblems
 120 with limited closed-form solutions, the offset wrist of the Kinova Gen3 Lite
 121 couples these objectives. Consequently, every waypoint becomes a fully cou-
 122 pled 6-DOF problem with multiple solution branches [17]. Solving these way-
 123 points independently offers no guarantee that adjacent solutions will remain
 124 on the same kinematic branch, inevitably leading to joint-space discontinu-
 125 ities along continuous paths.

Velocity-level controllers address this by operating directly in the differential domain. Resolved-rate motion control (RRMC) [18] computes joint velocities using the Jacobian pseudo-inverse ($\dot{\theta} = J^\dagger \cdot \dot{x}$). While damped least squares (DLS) [19, 20] introduce a damping factor λ to prevent velocity singularities ($\dot{\theta} = J^\top (JJ^\top + \lambda^2 I)^{-1} \dot{x}$). Quadratic velocity-level programming formulations [21] further impose joint limit constraints explicitly, offering safer hardware deployment [22]. These methods require an explicit, accurate Jacobian at every timestep. On non-spherical wrist manipulators, two further difficulties arise. First, position and orientation errors are coupled through all six rows of the Jacobian. Consequently, balancing translational and rotational accuracy requires complex, geometry-specific task-weighting matrices or damping schemes, and a setting tuned for one geometry need not transfer to another. Second, kinematic singularities no longer separate into distinct arm and wrist singularities, which makes the selection of a damping factor that is both safe near singular configurations and accurate away from them more difficult.

To improve upon the accuracy limitations of classical velocity controllers, hybrid approaches have emerged combining numerical methods with learned components [23]. Learning based methods has been explored as a residual corrector atop numerical solvers [24, 25], compensating for solver inaccuracies. An actor-critic RL approach was introduced as a correction signal in joint velocity space to the augmented PD controller for trajectory tracking on a 6-DOF industrial manipulator [26]. A gradient-free, model-free Bayesian Markov Chain Monte Carlo (MCMC) RL policy search utilized to circular trajectory control on 2-DOF manipulator in simulation to optimize the PD controller [27]. A Soft Actor Critic (SAC)-based outer-loop stochastic reference correction policy learned directly from the 7-DOF Baxter manipulator. Their approach augments an existing vendor provided hardware controller with bounded corrective actions as an inheritance rather than learning from the environment [28]. Further, [29] apply distributed PPO with LSTM and GAE to trajectory tracking on a 3-DOF SCARA robot and mobile robot, using random reference state initialization and an exponential position-error reward.

Further, a compound controller combining a Sliding Mode Controller (SMC) with SAC-based RL augmented by Lyapunov stability constraints is proposed for the trajectory tracking of manipulators [30, 31]. While their method requires a complete dynamic model of the system, it demonstrates that model-based initialization significantly improves RL sample efficiency. A

164 hybrid framework combining Twin Delayed Deep Deterministic Policy Gra-
 165 dient (TD3) with an RRT-generated reference trajectory is proposed [32].
 166 In this integrated planning and RL system, the reward function directly pe-
 167 nalizes deviations from the RRT-based reference, constraining the agent to
 168 refine the pre-computed joint-space path. Further, [33] formulate a collabo-
 169 rative PPO variant for tracking and vibration suppression of a flexible-joint
 170 manipulator, reporting that PPO-based policies converge faster and more
 171 stably than actor-critic and policy-gradient baselines under hardware-in-the-
 172 loop validation, a finding consistent with the convergence and smoothness
 173 advantages of clipped on-policy optimization observed in the present work.
 174 In a similar way, [34] proposed a traditional path planner-DRL framework,
 175 combining RRT* with the Tsallis actor-critic algorithm to find optimized-
 176 energy IK solutions for redundant manipulators. This approach treats IK as
 177 a pose-reaching problem, solved point-by-point using an ANN-based solver.

178 Several DDPG-based methods have been explored to enhance trajectory
 179 tracking, these often depend on simplified assumptions or known dynamics.
 180 [35] proposed a sequential DDPG approach for n-link manipulators, utilizing
 181 dual actor-critic networks trained progressively from 2D to 3D space to accel-
 182 erate convergence. Similarly, [36] implemented a distributed DDPG frame-
 183 work with prioritized experience replay, where multiple synchronous workers
 184 explore diverse policies to update a shared global network. Furthermore, [37]
 185 introduced hybrid variants, such as LC-DDPG and TD3-ADX, that learn
 186 joint torque commands by embedding an explicitly derived Euler-Lagrange
 187 dynamic model within the simulation. While these methods demonstrate
 188 efficacy in simulation, they frequently operate in joint torque space requir-
 189 ing precise manipulator dynamics, and lack real-time hardware validation,
 190 leaving the sim-to-real gap unaddressed. Moreover, these frameworks often
 191 overlook orientation control and motion smoothness, which are critical for
 192 scaling to 6-DOF manipulators.

193 The learning-based approaches discussed above fall into two distinct fam-
 194 ilies based on their output. The first family optimizes the motion path itself,
 195 generating an offline trajectory executed by a separate controller. Examples
 196 include RRT-guided frameworks [32, 34] and reinforcement-learning-driven
 197 path generation for musculoskeletal energy optimization [38]. The second
 198 family, which includes our proposed method, focuses on reactive differential
 199 IK. Here, the reference trajectory is supplied externally, and the controller
 200 maps instantaneous pose error and robot state directly to joint-velocity com-
 201 mands at each timestep. These approaches are complementary; an offline

202 learned path generator could seamlessly supply the reference tracked by an
 203 online reactive controller.

204 Despite these advances trajectory tracking, specifically on non-spherical
 205 wrist manipulators is unaddressed, IK solution introduces kinematic coupling
 206 that complicates both analytical solvers and standard RL reward formula-
 207 tions. Existing approaches either operate in torque space requiring explicit
 208 dynamic models, augment pre-existing classical controllers rather than learn-
 209 ing from scratch. Furthermore, no prior work demonstrates that a policy
 210 trained purely on static pose reaching over a joint limit aware reachability
 211 dataset, without any trajectory supervision can generalize to continuous tra-
 212 jectory tracking at deployment. The proposed framework addresses this gap
 213 directly: a compact 256×256 actor-critic architecture, trained end-to-end
 214 on this coupled reward, learns a generalizable velocity-level mapping that
 215 extends to the continuous trajectories evaluated here without requiring tra-
 216 jectory demonstrations, dynamic models, or task-specific re-training. The
 217 proposed method is therefore a reactive differential IK controller rather than
 218 a trajectory generator. It resolves joint velocities online for externally sup-
 219 plied reference paths, circumventing the need for an explicit Jacobian or the
 220 analytical position–orientation decoupling absent in non-spherical wrists.

221 3. Methodology

222 The proposed framework, illustrated in Fig. 2, describes our methodol-
 223 ogy for training the trajectory tracking task. The observation space $\mathbf{s}_t \in \mathbb{R}^{19}$
 224 provides the essential information required for the update of the policy. The
 225 action space $\mathbf{a}_t \in \mathbb{R}^6$ consists of six joint velocities sampled from the policy
 226 $\pi_\theta(\cdot|\mathbf{s}_t)$. The training environment, implemented in NVIDIA Isaac Lab, uti-
 227 lizes 4096 parallel environments to train actor-critic neural networks with a
 228 256×256 architecture. Further this setup uses Proximal Policy Optimiza-
 229 tion (PPO) to give efficient joint velocities for the efficient trajectory tracking
 230 explained below in further sections.

231 3.1. State and Action Space

232 The policy operates on a 19-dimensional observation vector as detailed in
 233 Table 1 and outputs 6-dimensional joint velocity commands $\dot{\boldsymbol{\theta}}_p$.

234 3.2. Simulation Environment

235 Training is conducted in NVIDIA Isaac Lab with a physics timestep of
 236 $\Delta t = 1/120$ s and a control decimation of 2, resulting in a policy frequency

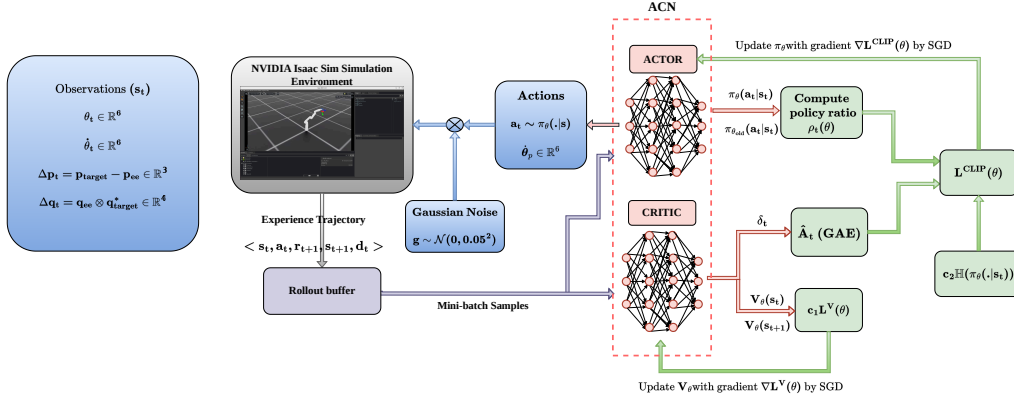


Figure 2: Overall Method for Differential Inverse Kinematics for Trajectory Tracking.

Table 1: Observation and Action Space

Type	Description	Symbol	Dim.
Observation	Joint positions	θ_t	6
	Joint velocities	$\dot{\theta}_t$	6
	Position error	$\Delta \mathbf{p}_t$	3
	Orientation error	$\Delta \mathbf{q}_t$	4
	Total observation		19
Action	Joint velocity commands	$\dot{\theta}_p$	6

237 of 60 Hz. The Kinova Gen3 Lite robot is modeled as a USD asset with self-
 238 collisions enabled, ensuring physically realistic interactions during training.
 239 The joints are velocity-controlled, enabling direct command of joint velocities
 240 from the policy output. The joint drive stiffness and damping parameters
 241 of the USD articulation were calibrated against the physical robot: constant
 242 joint-velocity commands were applied to the joints in simulation and on the
 243 Kinova Gen3 Lite, the resulting end-effector trajectories were recorded and
 244 compared, and the drive parameters were tuned until the simulated and
 245 measured trajectories matched. This calibration aligns the nominal actuator
 246 response of the simulation with that of the hardware. Each episode spans
 247 600 steps, with the robot initialized to its home configuration (zero position)
 248 at every reset. Episodes terminate only on time-out; no early termination on
 249 success, joint-limit contact, or collision is applied, so the policy is exposed to
 250 the full episode, including holding the pose after convergence. The reward
 251 contains no explicit joint-limit term: joint position limits are enforced by
 252 the PhysX articulation solver from the limits defined in the robot model,
 253 and joint-limit awareness enters training through the reachable-pose dataset,
 254 which is restricted to configurations within the soft joint limits (Section 3.3).
 255 At every episode reset, a target pose is drawn uniformly at random from
 256 the reachable-pose dataset described in Section 3.3, and the corresponding
 257 pose error ($\Delta \mathbf{p}_t$, $\Delta \mathbf{q}_t$ in Table 1) is computed relative to the robot’s current
 258 end-effector pose. Stochasticity during training is introduced not by per-
 259 turbing the target pose set itself, but at the policy output, via the additive
 260 Gaussian action noise of Eq. (1) (Section 3.4), which improves robustness to
 261 perturbations of the commanded joint velocities (Section 3.6). The training
 262 is conducted on an NVIDIA GeForce RTX 5060 Ti (16 GB VRAM) with the
 263 hyper parameters used are listed in Table 2.

264 3.3. Reachable-Pose Dataset Generation

265 The target poses used for training are drawn from a reachable-pose
 266 dataset generated once, offline, in NVIDIA Isaac Lab by exhaustive grid
 267 enumeration of the joint space (Fig. 3). Each of the six arm joints is dis-
 268 cretized into $N_s = 12$ uniformly spaced values over its soft position limits
 269 $[\theta_i^{\min}, \theta_i^{\max}]$, a resolution of $\Delta \theta_i = (\theta_i^{\max} - \theta_i^{\min}) / (N_s - 1) \approx 0.49$ rad ($\approx 28^\circ$).
 270 Their Cartesian product yields $N_s^6 \approx 3 \times 10^6$ candidate configurations, all
 271 joint-limit feasible by construction.

272 The candidates are evaluated in batches of 4096 parallel environments.
 273 In each environment, the robot is reset to its home configuration and driven

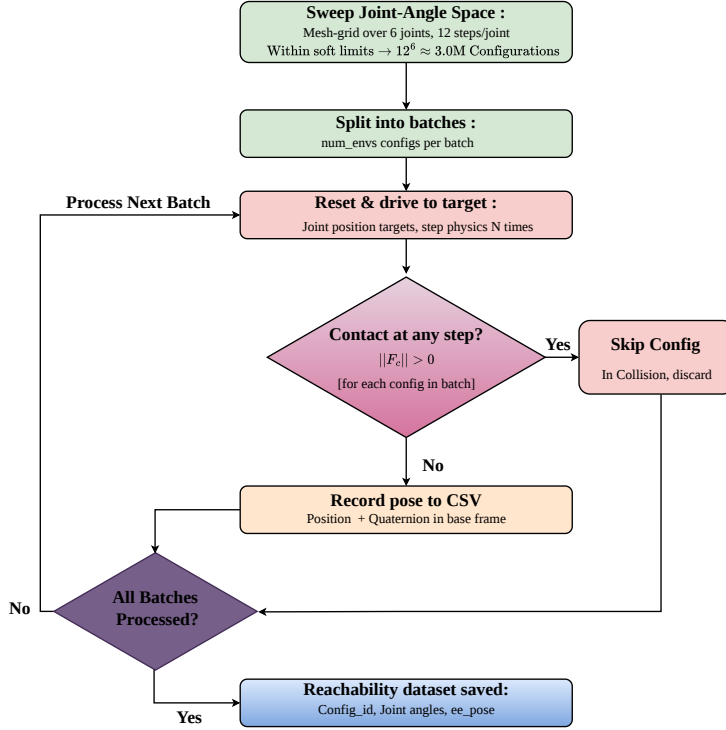


Figure 3: Offline generation of the collision-free reachable-pose dataset in Isaac Lab.

274 to the candidate configuration by joint position targets for 500 physics steps
 275 ($\Delta t = 1/60$ s), which is more than enough time to reach any candidate
 276 configuration, with the gripper held open. A contact sensor on every robot
 277 link monitors the net contact force throughout this motion, and a candidate
 278 is discarded if any link registers non-zero contact force at any step, i.e., self-
 279 collision or contact with the ground plane. For each retained configuration,
 280 the joint vector and the settled end-effector pose (position and quaternion)
 281 expressed in the robot base frame are stored. This yields approximately
 282 1.5 M collision-free reachable poses ($\approx 50\%$ of the candidates), which are
 283 reused unchanged throughout the 5×10^8 -step training run. Because targets
 284 are sampled on a joint-space grid, their Cartesian density is not uniform
 285 across the workspace. All evaluation trajectories are off-grid, so the tracking
 286 results in Section 4 reflect generalization between sampled targets rather than
 287 recall of them.

Table 2: PPO Hyperparameters

Hyperparameter	Value
Total Timesteps	5×10^8
Parallel Environments	4096
Steps per Rollout (N_{steps})	16
Minibatch Size	4096
Number of Epochs	10
Discount Factor (γ)	0.99
GAE Parameter (λ)	0.95
Learning Rate	5×10^{-5}
Clip Range (ϵ)	0.13
Entropy Coefficient (c_2)	0.01
Value Function Coefficient (c_1)	0.5
Max Gradient Norm	0.5
Network Architecture	$[256 \times 256]$
Activation Function	Tanh
Weight Initialization	Orthogonal

288 3.4. Rollout Collection and Policy Update

289 At each training iteration, experience tuples $\langle \mathbf{s}_t, \mathbf{a}_t, r_{t+1}, \mathbf{s}_{t+1}, \mathbf{d}_t \rangle$ are col-
 290 lected across all parallel environments over $N_{\text{steps}} = 16$ steps, yielding 65,536
 291 transitions per rollout. The policy $\pi_\theta(\cdot|\mathbf{s}_t)$ outputs a 6-dimensional joint ve-
 292 locity command $\boldsymbol{\theta}_p \in \mathbb{R}^6$, one per actuated joint of the Kinova Gen3 Lite. To
 293 improve robustness to real hardware actuator imprecision and to encourage
 294 exploration of the joint velocity space, additive Gaussian perturbation noise,
 295 $g \sim \mathcal{N}(0, 0.05^2)$ is injected at the environment interface before command
 296 execution as in Eq. (1):

$$\begin{aligned} \mathbf{a}_t &= \text{clip}(\pi_\theta(\cdot|\mathbf{s}_t) + g, -1, 1); \\ g &\sim \mathcal{N}(0, 0.05^2 \cdot \mathbf{I}_6) \end{aligned} \tag{1}$$

297 This perturbation serves a dual purpose, prevents the policy from overfitting
 298 to the noiseless simulation dynamics and [improves robustness to actuator-](#)
 299 [level perturbations of the commanded joint velocities.](#) It is not a substitute
 300 [for full domain randomization: latency, backlash, payload variation, and ac-](#)
 301 [tuator dynamics were not randomized in this work.](#) Advantage as in Eq. (2),
 302 estimates are computed via Generalized Advantage Estimation(GAE) using

303 the critic value $V_\theta(\mathbf{s}_t)$:

$$\begin{aligned}\hat{A}_t &= \sum_{l=0}^{N_{\text{steps}}} (\gamma\lambda)^l \delta_{t+l}; \\ \delta_t &= r_{t+1} + \gamma V_\theta(\mathbf{s}_{t+1}) - V_\theta(\mathbf{s}_t)\end{aligned}\tag{2}$$

304 with discount factor $\gamma = 0.99$ and GAE parameter $\lambda = 0.95$. The actor π_θ is
305 updated by maximizing the objective as in Eq. (3):

$$L(\theta) = L^{\text{CLIP}}(\theta) - c_1 \cdot L^V(\theta) + c_2 \cdot \mathbb{H}(\pi_\theta(\cdot|\mathbf{s}_t))\tag{3}$$

306 where

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

307 the policy ratio $\rho_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{old}}(a_t|s_t)$ measures the deviation from the
308 previous policy, the clipping range ϵ is 0.13, constrains the update step. The
309 critic is updated by minimizing mean squared error between the predicted
310 value and the target return $L^V(\theta) = \mathbb{E}_t[(V_\theta(s_t) - \hat{R}_t)^2]$, and $\hat{R}_t = \hat{A}_t + V_\theta(s_t)$ is
311 the target return, value function coefficient $c_1 = 0.5$, and entropy coefficient
312 c_2 is 0.01. Both actor-critic networks are parameterized with Tanh activations
313 and orthogonal initialization, updated jointly over 10 epochs per rollout via
314 minibatch SGD with batch size 4096 as listed in Table 2.

315 3.5. Reward Shaping

316 Designing a reward function for trajectory tracking on non-spherical wrist
317 manipulators requires careful balance between position and orientation error
318 convergence. The proposed reward function simultaneously minimizes both,
319 while enforcing progressive convergence and smooth joint motion. The pro-
320 posed reward function consists of three main components. [The structure of](#)
321 [this reward is dictated by the kinematics of the wrist. In spherical-wrist ma-](#)
322 [nipulators, disjoint joint groups decouple position and orientation, making](#)
323 [additive rewards adequate. Conversely, offset wrists couple these objectives:](#)
324 [every wrist joint simultaneously alters both position and orientation \[39\].](#)
325 [This coupling imposes three requirements on the reward formulation. First,](#)
326 [additive rewards are exploitable. A policy can maximize returns by con-](#)
327 [verging the better-conditioned error channel while neglecting the other. The](#)
328 [min operator \(Eq. \(6\)\) removes this incentive by basing the reward strictly](#)

on the larger normalized error, forcing simultaneous convergence. Second, fixed reward weights fail to generalize because error sensitivity varies across the workspace. Normalizing and min-coupling the errors eliminates manual weighting entirely. Third, kinematic coupling introduces multiple solution branches. Shifting between branches causes large joint displacements without reducing pose error. This is penalized by combining the asymmetric progress term (Eq. (9)) with the joint velocity penalty (Eq. (10)). The ablation study (Section 3.8) supports this account, although it does not isolate each requirement: the conventionally weighted variant fails on both channels, and the min-coupled variants converge on both channels once the joint velocity penalty is included (Table 4).

Pose Reward: The position and orientation errors are first normalized to a common scale:

$$\begin{aligned} e_p &= \frac{\|p_{\text{target}} - p_{\text{ee}}\|_2}{d_{\text{max}}}, \\ e_o &= \frac{2 \cdot \cos^{-1}(|\text{Re}(\mathbf{q}_{\text{ee}} \otimes \mathbf{q}_{\text{target}}^*)|)}{\pi} \end{aligned} \quad (4)$$

where d_{max} is the diagonal of the workspace, normalizing both errors ensures that the position and orientation errors are comparable in magnitude, preventing one from dominating the other. All quaternions use the scalar-first convention $[w, x, y, z]$ and are expressed in the robot base frame. The relative orientation $\Delta \mathbf{q}_t = \mathbf{q}_{\text{ee}} \otimes \mathbf{q}_{\text{target}}^*$ (Hamilton product) is provided directly to the policy as the orientation-error observation (Table 1). Since $\mathbf{q}$ and $-\mathbf{q}$ represent the same rotation, the orientation error e_o is computed from the absolute value of the scalar part of $\Delta \mathbf{q}_t$, which makes it invariant to the quaternion sign and bounds the rotation angle to $[0, \pi]$, so that $e_o \in [0, 1]$. All reported orientation metrics use the same sign-invariant angle, $2 \cos^{-1}(|\text{Re}(\Delta \mathbf{q}_t)|)$, in radians, i.e., before normalization by π .

The errors $\in \mathbb{R}$ as in Eq. (4), then shaped through a combined inverse-quadratic and logarithmic term as in Eq. (5):

$$r(x) = \frac{2.5}{1 + 100 \cdot x^2} - 0.1 \log(x + 10^{-8}) \quad (5)$$

where x is $e_{p/o}$.

The inverse-quadratic and logarithmic nature of the reward function helps to provide strong attraction in near-goal situations, and maintain a non-zero

358 gradient far from the goal, avoiding the problem of sparse rewards. The two
 359 terms are then coupled via a minimum operator as in Eq. (6):

$$r_{\text{pose}} = \min(r(e_p), r(e_o)) \quad (6)$$

360 The unconstrained algebraic combination of position and orientation rewards
 361 is susceptible to reward exploitation, where the agent converges to a subop-
 362 timal policy by maximizing one objective while leaving the other uncon-
 363 strained. The pose reward as shown in Eq. (6) applies the min operator to
 364 maintain simultaneous competency in both position and orientation through-
 365 out training.

366 **Progress Reward:** Although the coupled pose reward shapes the magni-
 367 tude of errors, it does not give an explicit incentive for the policy to reduce
 368 errors at every timestep. A policy could sustain a moderate error indefinitely
 369 and still receive a non-trivial reward. To address this, we introduce a progress
 370 reward as in Eq. (9), that explicitly penalizes stagnation and rewards per-
 371 step error reduction. At each timestep, the per-step error delta $\in \mathbb{R}$ as in
 372 Eq. (7), is computed for both position and orientation:

$$\Delta e_p = e_{p,t} - e_{p,t-1}, \quad \Delta e_o = e_{o,t} - e_{o,t-1} \quad (7)$$

373 where a negative delta indicates an improvement. Rather than applying
 374 progress rewards to both dimensions simultaneously, we apply it only to the
 375 dominant error, the dimension that is less in progress using the condition as
 376 in Eq. (8):

$$\Delta e = \begin{cases} \Delta e_p & \text{if } \Delta e_p \geq \Delta e_o \\ \Delta e_o & \text{if } \Delta e_p < \Delta e_o \end{cases} \quad (8)$$

377 The progress reward is then shaped asymmetrically to ensure a continuous
 378 gradient through zero:

$$r_{\text{prog}}(\Delta e) = \begin{cases} 4(-\Delta e + 0.25)^{0.5} - 2 & \Delta e < 0 \\ -2 \exp(2 \Delta e) + 2 & \Delta e \geq 0 \end{cases} \quad (9)$$

379 by offsetting the equations in Eq. (9), we prevent discontinuous reward cliffs
 380 that destabilize the value estimation. Square-root shaping on improvement
 381 provides diminishing returns for large progress steps, retarding aggressive
 382 overshooting. In contrast, the exponential penalty for regression imposes a
 383 sharply increasing penalty for error growth, creating a strong pressure on
 384 monotonic convergence.

385 **Joint Velocity Penalty:** Trajectory tracking demands not only accurate
 386 end-effector pose reaching, but also smooth, continuous joint movements
 387 throughout the trajectory tracking. To enforce smooth motion throughout
 388 the entire trajectory, we introduce a continuous **joint velocity penalty on the**
 389 **squared** joint velocity commands at every timestep as in Eq. (10):

$$r_{\text{vel}} = 1 - \exp \left(0.2 \sum_{i=1}^6 \dot{\theta}_i^2 \right) \quad (10)$$

390 the exponential formulation imposes a sharply increasing cost as the ag-
 391 gregate joint velocity increases. **This term acts as a control-effort (action)**
 392 **regularization on the commanded joint velocities; unlike energy-based for-**
 393 **mulations that model physical energy exchange, such as transactive energy**
 394 **in physical human-robot interaction [40], it does not represent mechanical**
 395 **energy or power consumption.**

396 The total reward is summation of pose, progression, and **joint velocity penalty**
 397 components as in Eq. (11).

$$r_{t+1} = r_{\text{pose}} + r_{\text{prog}} + r_{\text{vel}} \quad (11)$$

398 3.6. Sim-to-Real Transfer

399 Deploying the trained policy on physical hardware requires bridging the
 400 gap between simulation dynamics and real actuator behavior(see Fig. 4). At
 401 deployment, the policy output is clipped to $[-1, 1]$ before transmission to the
 402 Kinova Kortex API as in Eq. (12):

$$\dot{\theta}_p = \text{clip}(\mathbf{a}_t, -1, 1) \cdot \frac{180}{\pi} \quad [^\circ/\text{s}] \quad (12)$$

403 converting from normalized radians to degrees per second as required by the
 404 hardware interface. Observation parity between simulation and hardware is
 405 maintained directly from actuator feedback through the Kinova Kortex API,
 406 preserving the exact observation structure $\mathbf{s}_t \in \mathbb{R}^{19}$ seen during training.

407 The policy is deployed on an Intel NUC9 Extreme (16-core CPU, 32 GB
 408 RAM), achieving inference at ≈ 1.3 kHz. The Kinova Gen3 Lite operates with
 409 a feedback and command frequency of ≈ 40 Hz each, imposed by the Kinova
 410 Kortex API. Since feedback reading and joint speed command dispatch are
 411 executed sequentially, the effective closed-loop control frequency is 20 Hz,
 412 well within real-time control requirements.

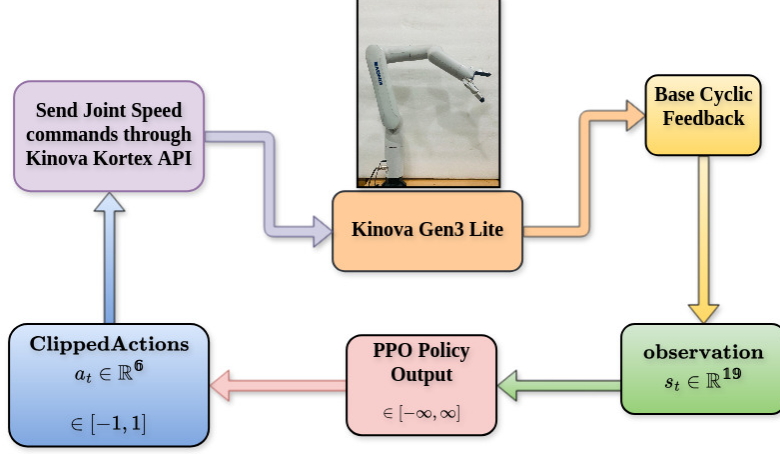


Figure 4: Sim-to-Real Architecture

413 This 20 Hz effective control frequency is approximately one-third of the
 414 60 Hz frequency used during training. Since the policy is a memoryless map-
 415 ping from instantaneous observation to velocity command with no recurrent
 416 state, a lower deployment frequency only extends the zero-order-hold (ZOH)
 417 duration of each command, it does not invalidate any individual policy in-
 418 put. The per-step Gaussian action noise (Eq. (1)) and joint velocity penalty
 419 (Eq. (10)) used during training jointly bound output velocity magnitude,
 420 limiting the inter-sample integration error introduced by a longer hold. Con-
 421 sistent with this, no frequency-related degradation was observed: hardware
 422 position RMSE on the circular trajectory (6.3 mm) was lower than simulation
 423 RMSE (9.95 mm) at the tested trajectory speed (0.1 rad/s), possibly aided
 424 by actuator damping and the low-level joint velocity controller smoothing the
 425 discretized commands, although this was not tested in isolation. This should
 426 be read as evidence against degradation at the tested speed, not as gen-
 427 eral frequency invariance; robustness at substantially higher tracking speeds,
 428 where pose error drifts more within a single 50 ms interval, remains untested
 429 and is left to future work.

430 The policy receives neither the desired Cartesian velocity nor any observa-
 431 tion history. Operating as a purely reactive state-feedback law, it computes
 432 actions based on instantaneous pose error without explicitly anticipating tar-
 433 get motion. Consequently, continuous tracking manifests as a sequence of

quasi-static pose-reaching tasks: at each control step, the reference is resampled and the policy drives the current error toward zero. Similar to classical resolved-rate motion control lacking a feedforward term [18], this error-driven feedback induces a bounded tracking lag proportional to v/K , where v is the Cartesian reference speed and K is the effective closed-loop gain. At the evaluated speed ($r\omega = 15$ mm/s), the reference advances only ≈ 0.25 mm per step at 60 Hz and ≈ 0.75 mm per step at 20 Hz. Because this spatial delta is negligible, each timestep closely mirrors the static reaching environment utilized during training. Furthermore, the asymmetric progress reward (Eq. (9)) strictly penalizes step-wise error growth, implicitly maximizing K to minimize steady-state lag. As trajectory speed increases, this lag grows proportionally with v until joint velocity commands saturate, at which point bounded tracking fails. We evaluate this lag scaling across 0.05–0.3 rad/s in Section 4.1 (Table 8). Characterizing the absolute speed limit and incorporating velocity feedforward or look-ahead horizons for anticipatory tracking remain as future work.

Joint velocity commands are bounded by hardware velocity limits enforced at the actuator level and further soft-constrained by the exponential joint velocity penalty in Eq. (10), which imposes a sharply increasing cost on aggregate joint speed. The hard clipping of policy outputs to $[-1, 1]$ before dispatch bounds the magnitude of every command at both the simulation and hardware interfaces. Clipping limits only command magnitude, however, and does not by itself guarantee safety near kinematic singularities, at joint limits, or under unexpected contact. In simulation, joint position limits are enforced by Isaac Lab’s physics solver rather than by the policy.

3.7. Baseline Comparison for Trajectory Tracking

To learn smooth and stable trajectory tracking, we rigorously evaluated existing DRL methods. Fig. 5 illustrates their mean terminal reward progression through training steps, normalized with respective maximum possible values to ensure fair comparison between differing reward structures. The sparse variants (suffix -S) replace the pose and progress terms with a tolerance signal of -1 per channel while the position or orientation error exceeds 4 cm or 0.1 rad, respectively, and 0 otherwise, plus a bonus of $+10$ when both are within tolerance; the joint velocity penalty (Eq. (10)) is retained. PPO-D was trained with three seeds (42, 100, and 200), which converge to the same plateau (Fig. 6); Fig. 5 shows seed 42.

470 The performance divergence between PPO and SAC highlights a critical
 471 trade-off between gradient stability and sample efficiency in trajectory track-
 472 ing. While SAC is widely regarded as a strong baseline for continuous control,
 473 PPO outperformed it in this task. One plausible explanation is PPO’s clipped
 474 surrogate objective, which restricts the magnitude of policy updates and may
 475 implicitly suppress high-frequency velocity fluctuations under the dense re-
 476 ward, although this mechanism was not isolated by a dedicated ablation.
 477 This is evident in Fig. 5, PPO-D achieves monotonic improvement, consis-
 478 tent with it tolerating the conflicting gradients introduced by joint velocity
 479 penalties and progression rewards, converging to a smooth control law with
 480 a mean terminal reward of ≈ 0.93 . Conversely, SAC-D substantially under-
 481 performs its sparse counterpart (SAC-S), which is consistent with—though
 482 not definitive proof of—an underlying conflict between the maximum-entropy
 483 objective, which encourages exploratory joint motion, and the joint velocity
 484 penalty, which incentivizes near-idle behavior to minimize control effort at
 485 the expense of task progress.

486 TRPO-D, despite sharing PPO’s conservative update logic through an
 487 exact KL-divergence constraint, converges slower and plateaus at a notably
 488 lower terminal reward ≈ 0.71 , suggesting that the computational overhead
 489 of enforcing tight trust region boundaries impedes timely policy refinement
 490 compared to PPO’s more tractable clipping approximation under dense re-
 491 ward supervision. PPO-S’s stagnation at ≈ 0.12 reveals that without dense
 492 reward shaping, the agent receives feedback too infrequently to learn the pre-
 493 cise joint coordination required for long-horizon differential IK trajectories.
 494 The sparse signal simply cannot distinguish between partially correct and
 495 fully correct motion sequences. Together, these results suggest that neither
 496 reward structure nor algorithm choice alone determines performance rather,
 497 it is their interaction dense rewards are necessary for precise credit assign-
 498 ment, and among the optimizers tested, PPO was the most robust to the
 499 conflicting gradients they introduce. For reactive trajectory tracking appli-
 500 cations where target poses are determined online rather than precomputed,
 501 path fidelity and steady-state precision are paramount, and PPO-D’s sta-
 502 ble, variance-controlled optimization proves more technically suited to this
 503 requirement than the alternative on-policy and off-policy baselines evaluated
 504 here.

505 Wall-clock training cost varies substantially across methods (Table 3).
 506 TRPO-D requires 3h 14m 36s to complete training, roughly $3\times$ PPO-D’s 1h
 507 5m 35s, consistent with the overhead of its exact KL-divergence constraint

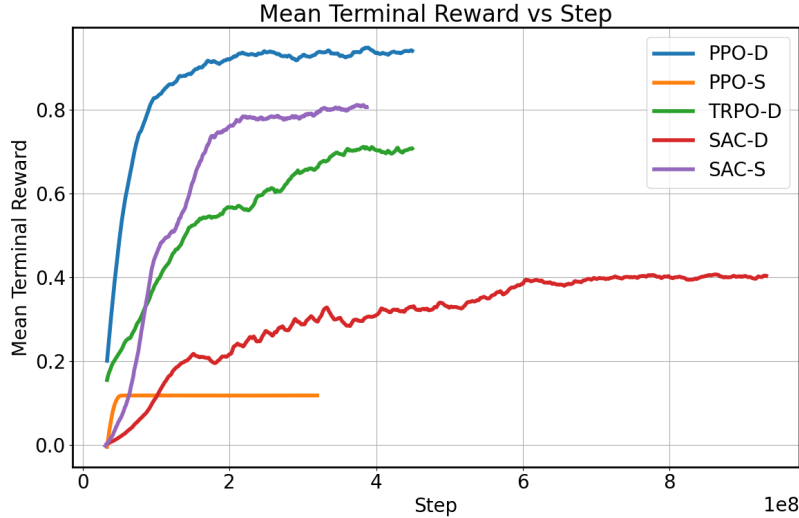


Figure 5: Training Curves for PPO Dense Reward, PPO Sparse Reward, TRPO Dense Reward, SAC Dense Reward, and SAC Sparse Reward

discussed above. SAC-D was trained for approximately 9×10^8 steps, nearly double SAC-S’s 4×10^8 step run, in the expectation that the dense reward would eventually outperform its sparse counterpart given sufficient additional experience; despite this extended budget, SAC-D’s terminal reward (Fig. 5) remains well below SAC-S’s, reinforcing rather than resolving the [entropy-velocity-penalty](#) conflict discussed above, and its 2h 52m 34s wall-clock cost reflects this longer run rather than a higher per-step cost. PPO-S terminates fastest (38m 52s) due to early stagnation detection. These figures should be read alongside [17]’s zero-cost closed-form analytical solution for the same arm: the proposed method’s accuracy and robustness gains come at a one-time, sub-2-hour training cost, not a per-deployment cost, and the policy requires no retraining across the four evaluated trajectory geometries.

3.8. Reward Component Ablation Study

The proposed reward function in Eq. (11), is validated to showcase the importance of each component in stable and smooth trajectory tracking. We evaluated four variants (a) Conventional, (b) [Min + Progress](#), (c) [Min + Velocity Penalty](#), and (d) Full (proposed), where “Min” denotes the [min-operator pose reward](#) (Eq. (6)). For ablation studies, the circular trajectory



Figure 6: PPO-D training curves for three seeds, min-max normalized over all three runs (scale not comparable to Fig. 5).

Table 3: Wall-clock training time and total timesteps per algorithm.

Method	Total Timesteps	Wall-clock Time
PPO-D (proposed)	$\approx 5 \times 10^8$	1h 05m 35s
PPO-S	$\approx 3.5 \times 10^8$	38m 52s
TRPO-D	$\approx 5 \times 10^8$	3h 14m 36s
SAC-D	$\approx 9 \times 10^8$	2h 52m 34s
SAC-S	$\approx 4 \times 10^8$	1h 24m 01s

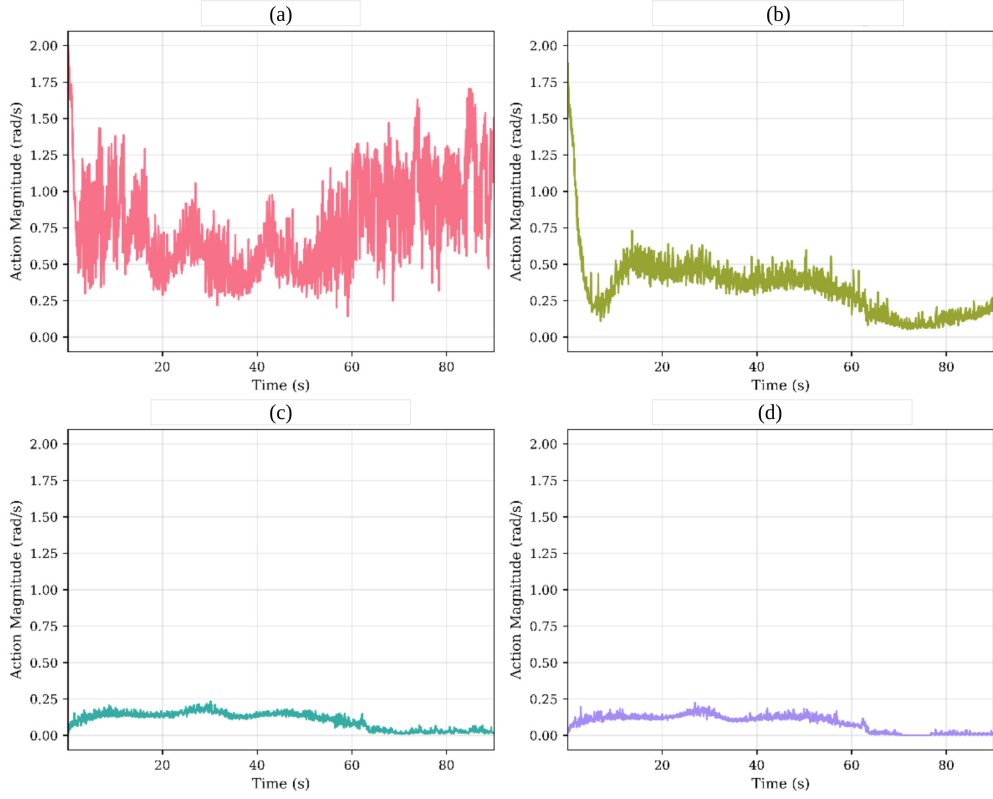


Figure 7: Joint Velocity Command Magnitudes across reward ablation variants during circular trajectory tracking: (a) Conventional, (b) **Min + Progress**, (c) **Min + Velocity Penalty**, (d) **Full (proposed)**. The exponential **joint velocity penalty** in variants (c) and (d) suppresses peak action magnitude relative to (a) and (b).

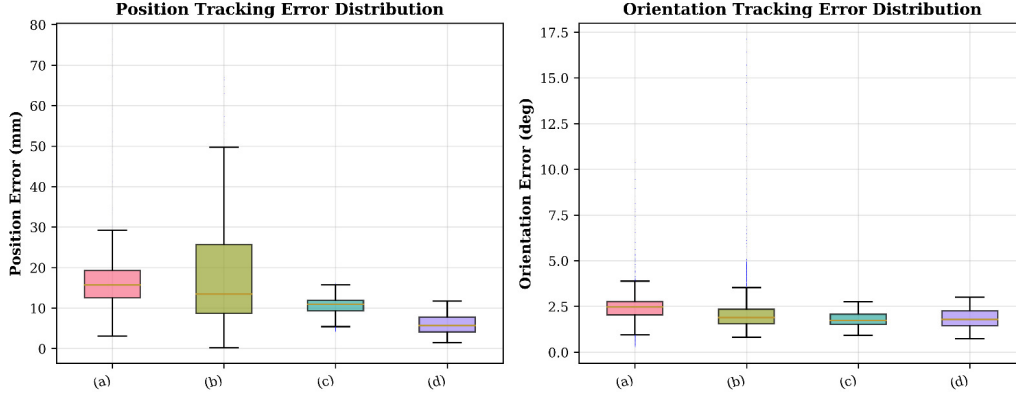


Figure 8: Position and Orientation Tracking Error Distribution across reward ablation variants on circular trajectory tracking. Variants (a)-(d) as in Fig. 7.

considered across all the listed variants, with summary metrics additionally reported on the 3-D Lissajous.

The exponential joint velocity penalty directly suppresses joint velocity magnitude, reducing peak actions from 2.0 rad/s in the conventional reward to below 0.25 rad/s in our full reward as shown in Fig. 7. The min-operator pose reward prevents the policy from exploiting high-velocity commands to satisfy one error dimension while neglecting the other. This coupling effect is visible in variant (b), where error progression without the joint velocity penalty produces large transient velocity spikes before collapsing into erratic oscillations throughout the trajectory.

From Fig. 8, the conventional reward (a) maintains persistent oscillations with a median error of 15–20 mm throughout, never settling below the threshold. Variant (c) achieves stable sub-threshold tracking but with a higher steady-state median of 10 mm. The full reward (d) achieves the lowest median position error of 6 mm with the tightest distribution, consistently tracking below the 10 mm threshold across the entire trajectory, demonstrating that all three reward components are necessary for robust and smooth tracking. In addition, these four variants for tracking circular trajectory as shown in Fig. 9, the conventional reward (a) fails to complete the circular path, deviating significantly near the starting point. The Variant (b) completes the circle but with visible radial deviation. Variants (c) and (d) closely follow the desired trajectory, with (d) achieving near-perfect

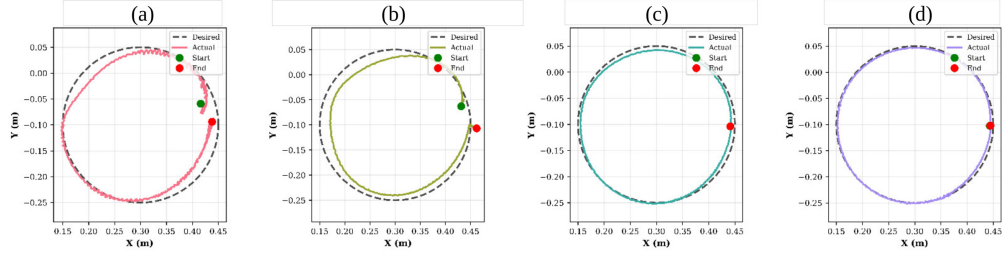


Figure 9: XY Trajectory comparison across reward ablation variants on circular trajectory tracking. Variants (a)-(d) as in Fig. 7.

circular tracking.

Table 4 summarizes all variants by RMSE on the circle and the 3-D Lissajous; for (a) and (b), the RMSE reflects the large deviations visible in Fig. 9, whereas the medians above describe their typical error. The ranking is preserved on the Lissajous: without the joint velocity penalty, (a) and (b) fail on both curves; adding it reduces position RMSE to 12–18 mm; and the progress term further improves position on both curves and orientation on the Lissajous (0.053 to 0.044 rad), where the orientation demand is higher. Variant (d) is PPO-D, trained with three seeds (Table 5); variants (a)–(c) are single-seed.

Table 4: Reward ablation on the circle and the 3-D Lissajous (position and orientation RMSE, smoothness index).

Variant	Circle			3-D Lissajous		
	Pos. (m)	Orn. (rad)	Smooth. (rad/s ²)	Pos. (m)	Orn. (rad)	Smooth. (rad/s ²)
(a) Conventional	0.1969	0.0800	50.31	0.2226	0.0858	32.71
(b) Min + Progress	0.1421	0.1045	28.81	0.1178	0.0818	32.89
(c) Min + Velocity Penalty	0.01223	0.0225	0.0055	0.01822	0.0533	0.0142
(d) Full (proposed)	0.00995	0.0265	0.0055	0.01667	0.0437	0.0147

4. Results and Discussion

This section presents experimental validation on the Kinova Gen3 Lite in both simulation and physical hardware. Section 4.1 compares the proposed

method against five classical and learning-based baselines in simulation across three trajectory geometries, circle, lemniscate, and 3-D Lissajous, chosen to span fixed-orientation, periodically-varying orientation, and high-curvature multi-axis motion respectively. In Section 4.2, evaluated sim-to-hardware transfer on a physical Kinova Gen3 Lite, where TRAC-IK, DLS, and SAC-S, one representative baseline per method family, are additionally deployed on the circular trajectory, while the proposed method is further evaluated on lemniscate, 3-D Lissajous, and square trajectories without retraining. All classical baselines retain the parameters reported in their respective foundational works, DLS with $\lambda = 0.05$ and Velocity QP with minimum regularization ($\lambda = 10^{-3}$); TRAC-IK uses the Distance solve type and is seeded at every waypoint with its own previous solution, the configuration most favorable for inter-waypoint continuity. Closed-loop configuration, per-method action scaling, and trajectory parameters common to both simulation and hardware trials are detailed in Section 4.1.

4.1. Simulation Result

All comparisons are carried out in the NVIDIA Isaac Sim simulation environment on circular, lemniscate, and 3-D Lissajous trajectories (Table 7), with end-effector overlays shown for circle and 3-D Lissajous in Fig. 10 and Fig. 11. All baselines and the proposed method operate under closed-loop Cartesian feedback: the velocity-level baselines (DLS, Velocity QP) recompute the Jacobian and pose error at every control timestep rather than integrating open-loop, and TRAC-IK is re-seeded with its previous solution at every waypoint. TRAC-IK is therefore evaluated with the standard seeding mitigation in place. No trajectory smoothing, lookahead filtering, or online (causal) filtering is applied to any method, so that all methods are compared on their unfiltered commands; lookahead methods would in any case require upcoming waypoints that the online setting considered here does not provide. Per-method action scales as in Table 7 were tuned on the circular trajectory to equalize each method’s native output magnitude to the robot’s velocity limits, then frozen and applied identically across all subsequent trajectories. No method, including the proposed policy, is provided with the desired Cartesian velocity of the reference; all methods compute their commands from the current reference pose and the measured robot state, so the comparison is made under identical reference information. SAC-S, which demonstrated superior training performance (Fig. 5), is utilized across all tracking evaluations. The primary metrics reported for the proposed method reflect training seed 42,

598 while comprehensive results for all three independent seeds are provided in
599 Table 5. Classical baselines are deterministic and evaluated once; TRPO-D
600 and SAC-S results represent single-seed policies. Complete baseline imple-
601 mentation parameters are detailed in Table 6, with TRAC-IK executed in
602 Gazebo via ROS 1 using identical reference trajectories.

Table 5: Simulation results of the proposed method for three training seeds (mean $\pm$ sample std).

Metric	Curve	Seed 42	Seed 100	Seed 200	Mean $\pm$ std
Pos. RMSE (m)	Circle	0.00995	0.00859	0.01403	0.01086 ± 0.00283
	Lemniscate	0.01065	0.01893	0.01601	0.01520 ± 0.00420
	Lissajous	0.01667	0.01723	0.01106	0.01498 ± 0.00341
Orn. RMSE (rad)	Circle	0.0265	0.0275	0.0296	0.0279 ± 0.0016
	Lemniscate	0.0235	0.0235	0.0349	0.0273 ± 0.0066
	Lissajous	0.0437	0.0460	0.0500	0.0466 ± 0.0032
Smoothness (rad/s ²)	Circle	0.0055	0.0037	0.0061	0.0051 ± 0.0012
	Lemniscate	0.0107	0.0057	0.0123	0.0096 ± 0.0034
	Lissajous	0.0147	0.0120	0.0184	0.0151 ± 0.0032

Table 6: Implementation of the classical baselines. DLS and Velocity QP share the pose error $\mathbf{e} = [\Delta \mathbf{p}; \phi \hat{\mathbf{u}}]$ and a Jacobian $\mathbf{J}$ obtained by finite differencing of the forward kinematics ($\delta = 10^{-4}$ rad), recomputed at every step. All commands are multiplied by the frozen a_{scale} of Table 7. For TRAC-IK, $\boldsymbol{\theta}$ denotes the measured joint positions, used only to convert the solution into a velocity command.

Method	Control law	Parameters
DLS	$\dot{\boldsymbol{\theta}} = \mathbf{J}^\top (\mathbf{J}\mathbf{J}^\top + \lambda^2 \mathbf{I})^{-1} \mathbf{e}$	$\lambda = 0.05$; unity gain
Velocity QP	$\min_{\dot{\boldsymbol{\theta}}} \frac{1}{2} \dot{\boldsymbol{\theta}}^\top (\mathbf{I} + \lambda \mathbf{J}^\top \mathbf{J}) \dot{\boldsymbol{\theta}} - (\mathbf{J}^\top \mathbf{e})^\top \dot{\boldsymbol{\theta}}$ s.t. $ \dot{\theta}_i \leq 1$ rad/s	$\lambda = 10^{-3}$; OSQP (tol. 10^{-4} , ≤ 1000 iter.); zero command if unsolved
TRAC-IK	Distance solve, then $\dot{\boldsymbol{\theta}} = (\boldsymbol{\theta}_{\text{IK}} - \boldsymbol{\theta})/\Delta t$, $\Delta t = 0.034$ s	$\epsilon = 10^{-5}$; 5 ms timeout; seeded with previous solution; held on failure

603 No single classical baseline is uniformly best, and the pattern of failure
604 for each is a direct consequence of its underlying mechanism rather than a
605 tuning artifact. Motion smoothness is quantified by the smoothness index
606 S as in Eq. (13), defined as the root-mean-square joint angular acceleration

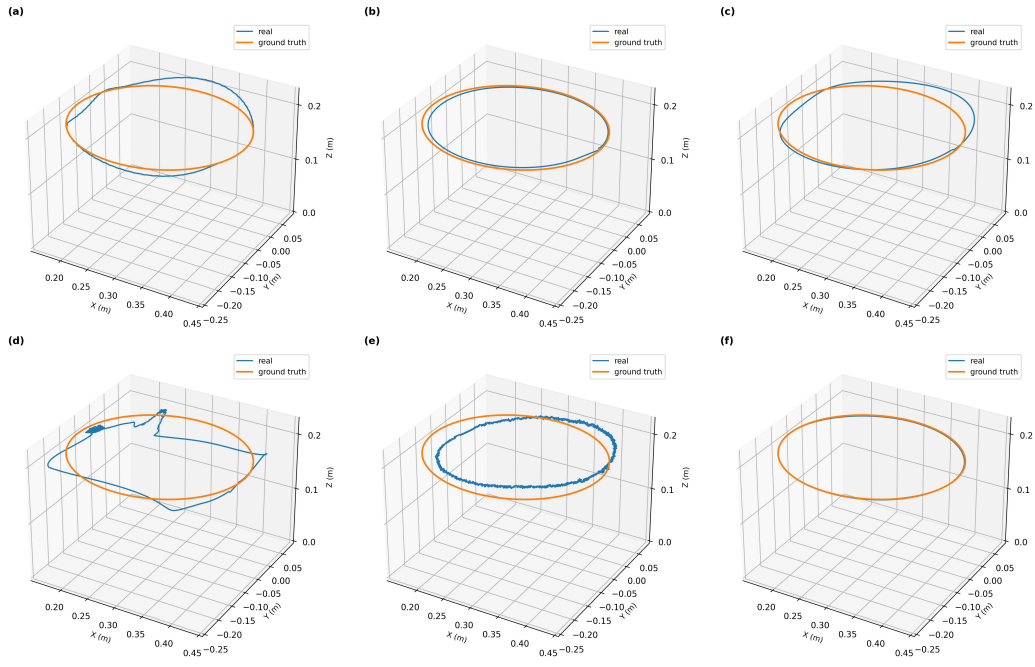


Figure 10: Circle Trajectory Comparison (a) TRAC-IK (b) DLS (c) Velocity QP (d) TRPO-D (e) SAC-S (f) PPO-D

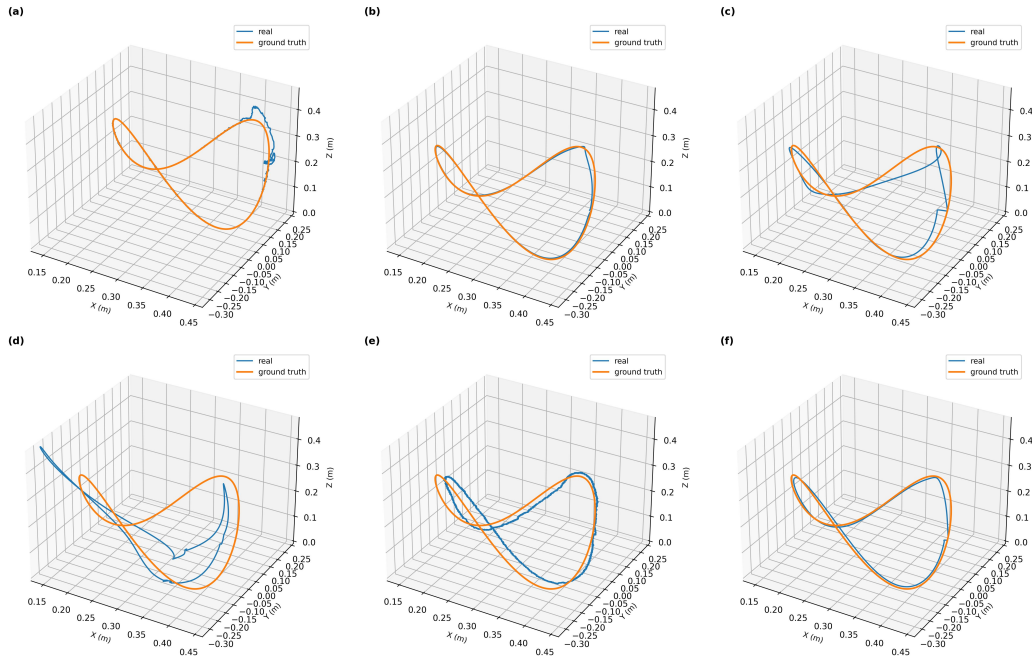


Figure 11: 3-D Lissajous Trajectory Comparison (a) TRAC-IK (b) DLS (c) Velocity QP (d) TRPO-D (e) SAC-S (f) PPO-D

Table 7: Simulation performance across circle, lemniscate, and 3-D Lissajous trajectories with frozen per-method action scales (Section 4.1). No classical baseline is uniformly best; **PPO-D** is the only method with stable accuracy and smoothness across all three geometries. Best value per metric in bold.

Method	Curve	a_{scale}	Pos. RMSE ↓ (m)	Orn. RMSE ↓ (rad)	Smoothness ↓ (rad/s ²)
TRAC-IK	Circle	0.2	0.001242	0.003409	54.547754
	Lemniscate	0.2	0.001073	0.004332	52.002110
	Lissajous	0.2	0.035272	0.052286	62.164647
DLS	Circle	1	0.016046	0.000767	0.013438
	Lemniscate	1	0.013554	0.036097	0.015920
	Lissajous	1	0.027939	0.020683	0.022242
Velocity QP	Circle	1	0.011013	0.002900	0.012409
	Lemniscate	1	0.013254	0.003857	0.029433
	Lissajous	1	0.035303	0.008647	0.016406
TRPO-D	Circle	0.4	0.089448	0.372891	2.918600
	Lemniscate	0.4	0.130428	0.383532	0.141100
	Lissajous	0.4	0.122329	0.257595	0.814877
SAC-S	Circle	0.2	0.029320	0.051513	19.778952
	Lemniscate	0.2	0.030142	0.051104	19.077041
	Lissajous	0.2	0.029181	0.050908	18.851007
PPO-D	Circle	1	0.009951	0.026476	0.005545
	Lemniscate	1	0.010650	0.023536	0.010719
	Lissajous	1	0.016667	0.043686	0.014745

607 computed from the joint velocities $\dot{\theta}_{i,k}$ logged at every control step of a trial:

$$S = \sqrt{\frac{1}{6(N-1)} \sum_{k=1}^{N-1} \sum_{i=1}^6 \left(\frac{\dot{\theta}_{i,k+1} - \dot{\theta}_{i,k}}{t_{k+1} - t_k} \right)^2} \quad [\text{rad/s}^2], \quad (13)$$

608 where N is the number of logged steps and i indexes the six joints. Lower S
609 indicates smoother joint motion.

610 TRAC-IK resolves each waypoint as an independent constrained opti-
611 mization, [seeded with its previous solution](#). In a noiseless simulation this
612 allows near-exact convergence at every individual pose, producing the lowest
613 position RMSE among all methods on the circle (1.2 mm) and lemniscate
614 (1.07 mm), Table 7. However, because each solve is independent with no
615 explicit velocity-domain coupling between consecutive timesteps, small dif-
616 ferences in optimizer convergence translate directly into joint-space disconti-
617 nuities frame-to-frame. [This is consistent with its smoothness index of 52–62,](#)
618 [the largest of any method: about three times that of SAC-S, 19–370 times](#)
619 [that of TRPO-D, and three to four orders of magnitude above DLS, Velocity](#)
620 [QP, and PPO-D](#). On the Lissajous trajectory, where curvature and orienta-
621 tion change, the nearest-seed assumption that keeps TRAC-IK well-behaved
622 on planar curves breaks down, and the solver begins to favor configuration
623 branches discontinuous with the previous solution, position RMSE degrades
624 by roughly $30\times$ (1.2 mm $\rightarrow$ 35.3 mm) relative to the circle.

625 DLS computes joint velocities from a damped pseudo-inverse Jacobian
626 with a fixed damping factor $\lambda = 0.05$ and a single, undifferentiated gain ap-
627 plied to the combined position-orientation error vector. On the circle, where
628 orientation is held constant, this fixed gain structure strongly favors orienta-
629 tion correction, since the magnitude of angular correction required to track
630 a fixed orientation is small and well within the controller’s effective band-
631 width, yielding the best orientation RMSE of any method (7.7×10^{-4} rad) at
632 the cost of a comparatively large 16 mm position error. On the lemniscate,
633 where orientation must vary continuously alongside position, the same fixed
634 gain cannot adaptively reallocate correction authority between the two error
635 channels, and orientation RMSE rises to 0.036 rad, worse than the proposed
636 method’s 0.024 rad on the identical curve. DLS’s strong circle orientation
637 result is therefore a constant-orientation artifact of its fixed-gain structure,
638 not a general advantage over learned methods.

639 Velocity QP [uses the same Jacobian and pose error as DLS but solves](#)
640 [a regularized quadratic program with explicit joint-velocity bounds at every](#)

step (Table 6). It attains the best orientation RMSE of any method on the lemniscate and the 3-D Lissajous (3.9×10^{-3} and 8.6×10^{-3} rad), but does not lead in position on any curve (11.0–35.3 mm), degrading on the Lissajous to a level comparable with TRAC-IK; this is consistent with the single fixed weighting of the combined pose error, shared with DLS, favouring the orientation channel.

TRPO shares PPO’s coupled reward (Eq. (6), Eq. (9), Eq. (10)) and its on-policy, trust-region-constrained update rule, but enforces an exact KL-divergence constraint rather than PPO’s clipped surrogate objective. The exact constraint requires a more expensive constrained optimization step per update; under an identical training budget (Table 3), this leaves TRPO’s policy less converged than PPO’s at the same total timestep count, producing both higher residual position/orientation error (8.9 cm / 0.37 rad on the circle) and substantially worse smoothness on the circle (2.92) than PPO (0.0055), since incomplete convergence leaves residual policy stochasticity in the action distribution.

SAC-S settles at steady-state errors of ≈ 2.9 cm and ≈ 0.051 rad across all three trajectories. These values lie within the 4 cm / 0.1 rad tolerance of its sparse reward, inside which the pose component of the reward is constant, so the policy has no incentive to reduce the error further; this is consistent with an offset independent of trajectory geometry. Its elevated smoothness index (18.9–19.8) may reflect the maximum-entropy objective, which the retained joint velocity penalty does not fully suppress within the tolerance zone; this was not isolated by a dedicated ablation.

PPO does not achieve the lowest position RMSE in simulation on every curve; rather, it is the most consistent method across geometries, with position RMSE confined to a 9.9–16.7 mm band (seed 42; 10.9–15.2 mm as three-seed means) across circle, lemniscate, and Lissajous, while every classical baseline degrades sharply on at least one curve (TRAC-IK by $\sim 30\times$ on Lissajous; DLS’s orientation by $\sim 47\times$ on lemniscate). PPO achieves the lowest mean smoothness index across all evaluated trajectories. It improves upon DLS and Velocity QP by a factor of 1.7–3 on the circle and lemniscate, performs comparably to Velocity QP on the Lissajous curve, and outperforms TRPO-D, TRAC-IK, and SAC-S by one to four orders of magnitude.

This consistency is in line with the structure of the proposed reward, although reward and optimizer were not varied independently for every baseline. The min-operator coupling in Eq. (6) prevents the policy from trading orientation accuracy for position accuracy under the shifting demands of each

679 curve, which is consistent with PPO not exhibiting the orientation collapse
 680 seen in DLS on the lemniscate, where position and orientation requirements
 681 vary continuously rather than remaining fixed as on the circle. The asym-
 682 metric progress shaping in Eq. (9) penalizes stagnation at every timestep
 683 regardless of trajectory geometry, consistent with PPO avoiding the persis-
 684 tent in-tolerance offset exhibited by SAC-S, which corresponds to sustained,
 685 under-penalized error under a sparser reward. The exponential joint velocity
 686 penalty in Eq. (10) acts directly on joint velocity magnitude, which plausibly
 687 underlies PPO’s smoothness advantage over every baseline, independent of
 688 trajectory shape. Neither the classical baselines nor the evaluated SAC-S use
 689 this reward, which may explain why their performance is uneven across ge-
 690 ometries; TRPO-D uses the same reward but, as noted below, converges to a
 691 lower training reward. The evidence supports the reward as necessary rather
 692 than sufficient: with the same optimizer, the sparse reward fails (PPO-S),
 693 and with the same reward and training budget, TRPO-D converges to a lower
 694 reward (Section 3.7); the SAC comparison additionally differs in reward and
 695 training budget. The result therefore reflects the proposed reward combined
 696 with a stable on-policy optimizer.

697 Table 8 reports the proposed method at reference angular velocities of
 698 0.05–0.3 rad/s (seed 42, simulation). Tracking remains bounded up to three
 699 times the nominal speed, and position RMSE grows approximately linearly
 700 with speed, roughly doubling from 0.1 to 0.3 rad/s on every curve, consistent
 701 with the tracking lag scaling as v/K (Section 3.6); the smoothness index rises
 702 more steeply, as faster references require larger joint corrections.

Table 8: Effect of reference angular velocity ω on the proposed method (seed 42, simulation).

Metric	Curve	ω (rad/s)			
		0.05	0.1	0.2	0.3
Pos. RMSE (m)	Circle	0.00829	0.00995	0.01431	0.01894
	Lemniscate	0.00837	0.01065	0.01609	0.02152
	Lissajous	0.01261	0.01667	0.02603	0.03529
Orn. RMSE (rad)	Circle	0.0216	0.0265	0.0369	0.0471
	Lemniscate	0.0183	0.0235	0.0377	0.0522
	Lissajous	0.0340	0.0437	0.0641	0.0833
Smoothness (rad/s ²)	Circle	0.0017	0.0055	0.0188	0.0387
	Lemniscate	0.0032	0.0107	0.0359	0.0720
	Lissajous	0.0042	0.0147	0.0505	0.1008

4.2. Real-time Results

The same circular, square, lemniscate, and 3-D Lissajous trajectories used in simulation (Section 4.1) are evaluated on hardware. All trajectories execute at 20 Hz closed-loop frequency, using the same trajectory shapes, sizes, and speed as in simulation ($r = 0.15$ m; side 0.2 m for the square, angular velocity 0.1 rad/s), centered at $[0.3, -0.1, 0.2]$ m. TRAC-IK, DLS, and SAC-S are validated on hardware for the circular trajectory (Fig. 12), one representative baseline per method family (classical numerical, classical velocity-level, and RL), while the proposed method is further evaluated on lemniscate, 3-D Lissajous, and square trajectories (Fig. 13, Fig. 14). Velocity QP is omitted from the hardware comparison because it shares the Jacobian, pose error, and fixed-weight structure of DLS, which therefore represents the classical velocity-level family on hardware. The error metrics observed of this study are listed in Table 9.

The sim-to-hardware transition reveals which methods rely on assumptions that do not survive contact with real actuator dynamics. TRAC-IK’s position RMSE degrades by roughly $29\times$ on hardware relative to simulation (1.2 mm $\rightarrow$ 36.2 mm), the largest gap of any method. This is consistent with its lack of a feedback term linking consecutive solves: each waypoint is resolved independently from joint encoder feedback with no mechanism to detect or correct accumulated real-world drift, transmission backlash, or actuator lag between solves, errors that do not exist in a noiseless simulation. DLS degrades by a comparatively moderate $\sim 3\times$ (16 mm $\rightarrow$ 49 mm), consis-

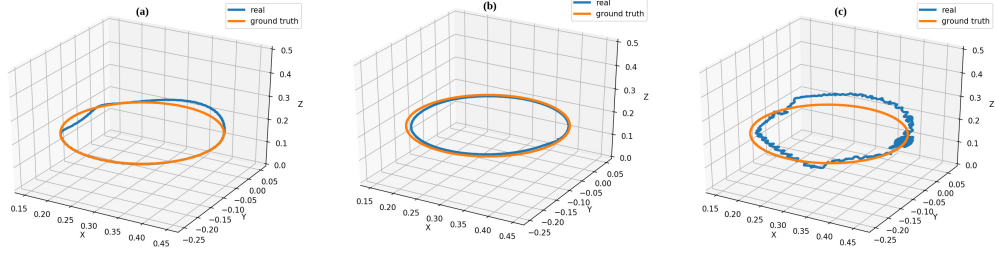


Figure 12: Real-time Deployment of (a) TRAC-IK (b) DLS (c) SAC-S

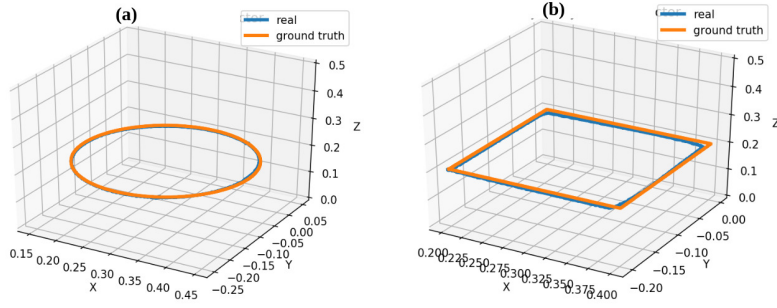


Figure 13: Real-Time Trajectory tracking of (a) Circle (b) Square with constant end effector orientation.

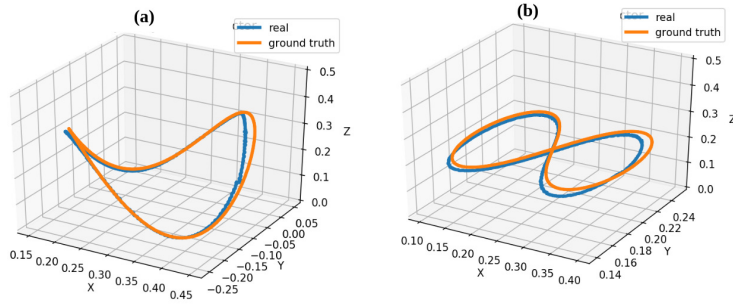


Figure 14: Real-Time Trajectory tracking of (a) 3-D Lissajous (b) Lemniscate with varying end effector orientation.

Table 9: Hardware performance on the Kinova Gen3 Lite: circle for all four methods, plus PPO-D generalization to square, lemniscate, and 3-D Lissajous without retraining. Best value per metric in bold.

Method	Curve	Pos. RMSE ↓ (m)	Orn. RMSE ↓ (rad)
TRAC-IK	Circle	0.0362	0.0770
DLS	Circle	0.0490	0.0021
SAC-S	Circle	0.0450	0.0590
PPO-D	Circle	0.0063	0.0320
	Square	0.0079	0.0334
	Lemniscate	0.0124	0.0212
	Lissajous	0.0154	0.0410

tent with its damping factor being calibrated against an idealized simulated Jacobian that does not account for joint compliance and backlash present on the physical Kinova Gen3 Lite; its orientation advantage on the circle persists on hardware (2.1×10^{-3} rad, still the best of any method), since the fixed-orientation condition that produces this advantage in simulation is preserved on hardware. SAC-S degrades by $\sim 1.5 \times$ (29 mm $\rightarrow$ 45 mm); its in-tolerance offset from simulation persists, and the additional error is attributed to unmodeled hardware effects.

On the circle, the only trajectory run by every method on hardware, PPO is the only method whose position error did not increase from simulation to hardware: its position RMSE is 6.3 mm on hardware against 9.95 mm in simulation. We attribute this transfer to several factors acting together rather than to any single mechanism: (i) the policy is a closed-loop reactive controller that re-reads the measured robot state at every step, so errors arising from residual model mismatch are corrected continuously rather than accumulated; (ii) the action and observation interfaces are identical in simulation and on hardware (joint-velocity commands to velocity-controlled actuators, with the same 19-D observation), and the simulated joint drives were calibrated against the physical robot (Section 3.2); (iii) the additive Gaussian action noise used during training (Eq. (1)) makes the policy robust to perturbations of the commanded velocities; and (iv) the evaluated regime of low-speed, unloaded, non-contact tracking keeps the effects of latency, backlash, and payload small. The policy also tolerated a threefold reduction in

control frequency from training (60 Hz) to hardware (20 Hz) without re-training, which provides partial evidence of robustness to timing mismatch. Because latency, backlash, payload variation, and actuator dynamics were not randomized, this result is established for the tested operating regime and should not be extrapolated beyond it without further robustness evaluation.

Across all four hardware trajectories, PPO maintains position RMSE between 6.3 mm and 15.4 mm without retraining (Table 9), with the lemniscate and 3-D Lissajous curves, which present orientation variation and curvature not explicitly represented in the static-pose training distribution, showing only a modest increase in error. This generalization is consistent with training on a dense, joint-limit-aware sample of the reachable workspace pose-error space (Section 3.3) rather than on trajectory-specific demonstrations: since the policy’s input is instantaneous pose error rather than a trajectory index, any sufficiently dense coverage of that error space at training time transfers to trajectory geometries never seen during training.

Taken together, the simulation and hardware results are consistent with the proposed reward structure, in combination with a stable on-policy optimizer, underlying the method’s accuracy-smoothness balance in simulation and its resistance to sim-to-real position-error degradation on hardware within the tested operating regime. Each baseline’s hardware result degrades by an amount consistent with a documented limitation of its control structure, absence of inter-waypoint feedback for TRAC-IK, fixed-gain miscalibration for DLS, persistent tolerance-band offset for SAC-S.

5. Conclusions

This work presented a reactive, online differential inverse kinematics controller for non-spherical wrist manipulator, in which a novel coupled reward function, combining a min-operator pose reward (Eq. (6)), asymmetric progress shaping (Eq. (9)), and an exponential joint velocity penalty (Eq. (10)), constitutes the central contribution rather than the choice of reinforcement learning algorithm. Operating on a compact 19-dimensional observation of joint state and instantaneous pose error (Table 1), the policy requires no explicit Jacobian inversion, no online IK solve, and no trajectory-specific supervision, learning purely from joint-limit-aware static pose-reaching samples (Section 3.3) and generalizing zero-shot to circular, square, lemniscate, and 3-D Lissajous trajectory tracking at deployment.

784 The simulation and hardware results (Section 4) show that no single clas-
 785 sical baseline is uniformly competitive across the evaluated metrics: TRAC-
 786 IK achieves the lowest position RMSE in simulation on planar curves but
 787 is less smooth than every other method, by up to four orders of magnitude,
 788 and degrades most severely on hardware (Table 9); DLS attains the best
 789 orientation tracking on the circle, but this advantage is specific to constant-
 790 orientation motion and is lost on the lemniscate; and SAC-S settles at a per-
 791 sistent offset within the tolerance of its sparse reward. The proposed method
 792 does not achieve the lowest simulation error on every individual curve, but
 793 it is the only method whose position RMSE remains within a narrow band
 794 across all three simulated geometries, whose mean smoothness is the best
 795 of any method on every curve, and whose hardware position error on the
 796 circle did not exceed its simulation error in the trials conducted, whereas
 797 that of every hardware baseline increased (Section 4.2). These properties
 798 are consistent with the proposed reward being necessary for the observed
 799 accuracy–smoothness–robustness trade-off, in combination with a stable on-
 800 policy optimizer, rather than attributable to the reward alone.

801 The present framework has several limitations that bound the scope of
 802 these claims. Hardware validation is restricted to a single manipulator, the
 803 Kinova Gen3 Lite; generalization to other non-spherical wrist platforms (such
 804 as KUKA, ABB, or UR variants with offset wrist axes) has not been tested,
 805 and the cost of porting the policy to a different kinematic structure is un-
 806 known. The TRPO-D and SAC-S baselines, along with ablation variants
 807 (a)–(c), were trained using a single seed. Furthermore, all physical experi-
 808 ments represent single-trial executions; consequently, hardware-level run-to-
 809 run variance remains uncharacterized. The trajectory speed evaluation is
 810 restricted to simulation using a single training seed. Furthermore, the effects
 811 of varying control frequencies and training action noise were not ablated;
 812 characterizing these parameters is left to future work. The empirical evalua-
 813 tion is bounded by specific operational constraints: four trajectory geometries
 814 executed at a single spatial scale within a localized workspace region, orig-
 815 inating from a fixed initial configuration. Reference speeds were evaluated
 816 between 0.05–0.3 rad/s in simulation and fixed at 0.1 rad/s on hardware.
 817 Generalization to arbitrary spatial scales, broader workspace regions, or ran-
 818 domized initial states remains untested. Hardware baseline comparison was
 819 restricted to the circular trajectory for TRAC-IK, DLS, and SAC-S, while
 820 the proposed method was further validated on lemniscate, 3-D Lissajous, and
 821 square trajectories without retraining; a head-to-head hardware comparison

822 on the geometrically more complex trajectories remains open. Controller be-
 823 havior near kinematic singularities also remains uncharacterized: although
 824 the robot is initialized at the zero configuration at every episode reset, the
 825 policy has not been explicitly trained on singularity-proximal configurations,
 826 and its recovery behavior at kinematic boundaries is therefore implicit rather
 827 than verified. Finally, because the policy has no explicit dynamic model or
 828 closed-form stability guarantee, it does not offer the formal safety guarantees
 829 available to model-based controllers, a property that may matter for contact-
 830 rich or safety-critical deployment even though it is not required for the re-
 831 active, non-contact tracking tasks evaluated here. [Energy-based approaches](#)
 832 [such as passivity-based control, which monitor the energy exchanged with](#)
 833 [the environment to maintain stable and safe physical interaction \[41\], offer](#)
 834 [such guarantees, whereas velocity clipping in the proposed method bounds](#)
 835 [only command magnitude and does not ensure safety at joint limits or under](#)
 836 [unexpected contact. In addition, latency, backlash, payload variation, and](#)
 837 [actuator dynamics were not randomized during training, so the sim-to-real](#)
 838 [result is established for low-speed, unloaded, non-contact tracking only.](#)

839 Future work will address these limitations directly. Singularity-proximal
 840 configurations will be incorporated during rollout collection to enable ver-
 841 ified recovery behavior at kinematic boundaries. Multi-seed training will
 842 be extended to the TRPO-D and SAC-S baselines and the reward-ablation
 843 variants, and the hardware baseline comparison will be extended to the non-
 844 circular trajectories. [The reference velocity or a short preview of upcoming](#)
 845 [waypoints will be added to the observation to enable anticipatory tracking,](#)
 846 [and the range of trajectory speeds over which the reactive policy maintains](#)
 847 [bounded error will be characterized. Domain randomization over latency,](#)
 848 [backlash, payload, and actuator parameters, together with an external safety](#)
 849 [layer such as a control-barrier-function or passivity-based safety filter \[41\],](#)
 850 [will be investigated to extend transfer and safety guarantees beyond the](#)
 851 [tested regime.](#) Validation on additional non-spherical-wrist manipulators will
 852 be pursued to test whether the consistency observed here is specific to the
 853 Kinova Gen3 Lite or a more general property of the reward design. Finally,
 854 the framework will be extended to dual-arm manipulation, where the learned
 855 differential IK controller would serve as a per-arm low-level policy for coordi-
 856 nated bi-manual control, a natural extension toward collaborative assembly
 857 operations requiring simultaneous precise trajectory tracking under contact
 858 constraints.

859 CRediT authorship contribution statement

860 Surya Prakash S.K: Writing – original draft, Conceptualization, Validation,
861 tion, Methodology, Formal analysis, Software, Investigation, Visualization,
862 Data curation. Sai Teja Majji: Writing – review and editing, Conceptu-
863 alization, Validation, Methodology, Formal analysis, Software. Darshanku-
864 mar Prajapati: Writing – review and editing, Conceptualization, Validation,
865 Methodology. Jagannath Prasad Sahoo: Writing – review and editing, Visu-
866 alization, Validation. Pushkar Kumar: Writing – review and editing, Visu-
867 alization, Validation. Jayakumar V.K: Writing – review and editing, Visual-
868 ization, Validation. Amit Shukla: Writing – review and editing, Supervision,
869 Conceptualization, Funding acquisition.

870 Declaration of competing interest

871 The authors declare that they have no known competing financial inter-
872 ests or personal relationships that could have appeared to influence the work
873 reported in this paper.

874 Data availability

875 The source codes used in this work are in this GitHub repository and the
876 data used for training is available in this drive.

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